

## *Acute dietary nitrate supplementation improves cycling time trial performance.*

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**PURPOSE:** Dietary nitrate supplementation has been shown to reduce the O<sub>2</sub> cost of submaximal exercise and to improve high-intensity exercise tolerance. However, it is presently unknown whether it may enhance performance during simulated competition. The present study investigated the effects of acute dietary nitrate supplementation on power output (PO), VO<sub>2</sub>, and performance during 4- and 16.1-km cycling time trials (TT). **METHODS:** After familiarization, nine club-level competitive male cyclists were assigned in a randomized, crossover design to consume 0.5 L of beetroot juice (BR; containing ~ 6.2 mmol of nitrate) or 0.5 L of nitrate-depleted BR (placebo, PL; containing ~ 0.0047 mmol of nitrate), ~ 2.5 h before the completion of a 4- and a 16.1-km TT. **RESULTS:** BR supplementation elevated plasma [nitrite] (PL = 241 ± 125 vs BR = 575 ± 199 nM,  $P < 0.05$ ). The VO<sub>2</sub> values during the TT were not significantly different between the BR and PL conditions at any elapsed distance ( $P > 0.05$ ), but BR significantly increased mean PO during the 4-km (PL = 279 ± 51 vs BR = 292 ± 44 W,  $P < 0.05$ ) and 16.1-km TT (PL = 233 ± 43 vs BR = 247 ± 44 W,  $P < 0.01$ ). Consequently, BR improved 4-km performance by 2.8% (PL = 6.45 ± 0.42 vs BR = 6.27 ± 0.35 min,  $P < 0.05$ ) and 16.1-km performance by 2.7% (PL = 27.7 ± 2.1 vs BR = 26.9 ± 1.8 min,  $P < 0.01$ ). **CONCLUSIONS:** These results suggest that acute dietary nitrate supplementation with 0.5 L of BR improves cycling economy, as demonstrated by a higher PO for the same VO<sub>2</sub> and enhances both 4- and 16.1-km cycling TT performance.